INDUSTRIAL ASSIGNMENT- 1 Project Initiation
STUDENT INTERNSHIP AND PLACEMENT MANAGEMENT PORTAL

1. Project Title and Overview

Project Title: Student Internship and Placement Management Portal

The Student Internship and Placement Management Portal is a centralized, web-based platform designed to manage the complete internship and campus placement cycle of a diploma college. It brings students, recruiting companies, the Placement Officer (Training and Placement cell), Faculty Coordinators and the Administrator onto one system so that profiles, opportunities, applications, interviews, offers and reports are handled in a single, secure and transparent place.

The project is carried out as a series of twelve Industrial Assignments. This document, Industrial Assignment 1, initiates the project by defining why it is needed, what it will deliver, who will use it, how it will be built and how its progress and success will be measured. All later assignments (Requirements Gathering, User Stories, Database Queries, ER Diagram, UI Design in Figma, Spring Boot development and Postman testing) are built on the foundation laid here.

2. Background and Problem Statement

2.1 Background

Every year, final-year diploma students look for internships and jobs through the college's Training and Placement (T&P) cell. In most colleges this process still depends on notice boards, spreadsheets, e-mail threads and messaging groups. Student details are scattered across many files, company requirements are communicated late, and the T&P cell spends a large part of the placement season on manual coordination instead of guiding students.

2.2 Problem Statement

There is no single, reliable system that allows students to discover opportunities, apply online and track results; allows companies to publish openings and manage candidates; and allows the T&P cell and faculty to monitor the whole process. As a result:

- Students miss opportunities because information reaches only a few of them or arrives after the deadline.
- Students repeatedly submit the same personal details and resume to different companies.
- Eligibility (CGPA, branch, semester) is checked manually, which is slow and error-prone.
- Companies cannot easily view, shortlist or communicate with suitable candidates.
- Interview schedules and offer decisions are shared informally and are often missed or misunderstood.
- There is no accurate, up-to-date record of how many students applied, were shortlisted, selected or placed.
- Access to student data is not controlled by role, which creates privacy and security risks.

3. Project Vision and Mission

Vision: To make every internship and placement opportunity visible, accessible and fairly managed for every eligible student of the college.

Mission: To provide a simple, secure and role-based digital platform that automates the internship and placement workflow from registration to offer acceptance and reporting.

4. Project Objectives

1. Provide secure registration, login, logout and access management for all user roles.
2. Allow students to create and maintain a complete academic profile and upload their resume once.
3. Allow recruiting companies to create a company profile and post, edit and close internship and placement opportunities.
4. Allow students to apply to opportunities online only when they meet the eligibility rules, and to track and withdraw applications.
5. Allow recruiters and the Placement Officer to shortlist, reject or select applicants and to maintain a status history.
6. Allow interviews to be scheduled and offers to be released, viewed, accepted or rejected through the portal.
7. Provide role-specific dashboards, summaries and exportable reports for placement monitoring.
8. Provide fast search and filter options for opportunities, students and applications.
9. Protect student and company data through role-based access control and secure password storage.

5. Scope of the Project

5.1 In Scope

- Five user roles with role-based access: Student, Company Recruiter, Placement Officer, Faculty Coordinator and Administrator (a Guest is an unregistered visitor of the login/registration pages).
- Seven functional modules (epics) covering 28 tasks, listed in Section 7.
- Internship and placement (full-time job) opportunities.
- Application lifecycle: Applied, Shortlisted, Interview Scheduled, Selected, Rejected and Withdrawn.
- Offer lifecycle: Pending, Accepted and Rejected.
- Maintenance of history for access changes, profile edits, opportunity edits and deletions, application status changes and report generation.

- Responsive web interface designed in Figma; REST API backend developed with Spring Boot; MySQL database; API testing with Postman.

5.2 Out of Scope (Current Version)

- Online aptitude tests, coding tests and video-interview facility inside the portal.
- Payment processing, stipend disbursement and legal offer-letter generation.
- Native mobile applications and alumni network features.
- Automatic resume parsing and AI-based candidate recommendation.
- Integration with the college's examination or ERP system (academic data is entered by the student and verified by the Placement Officer).

6. Stakeholders and User Roles

Stakeholder / Role	Description	Main Responsibilities in the System
Guest	A person who has not registered or logged in.	Can view the login and registration pages only; cannot access any protected feature.
Student	A diploma student of the college looking for internships or jobs.	Register, build profile, upload resume, search and apply to opportunities, track applications, view interviews, respond to offers.
Company Recruiter	HR or recruitment representative of a visiting company.	Create company profile, post and manage opportunities, review applicants, update status, schedule interviews, release offers.
Placement Officer	Head of the Training and Placement cell.	Approve recruiter accounts, verify student details, oversee all opportunities and applications, handle exceptions and generate placement reports.
Faculty Coordinator	Department-level coordinator or HoD monitoring placements.	Read-only monitoring of department students, summaries and reports.
Administrator	Technical administrator of the portal.	Manage user accounts, roles and access, deactivate accounts, restore deleted records and oversee the whole system.

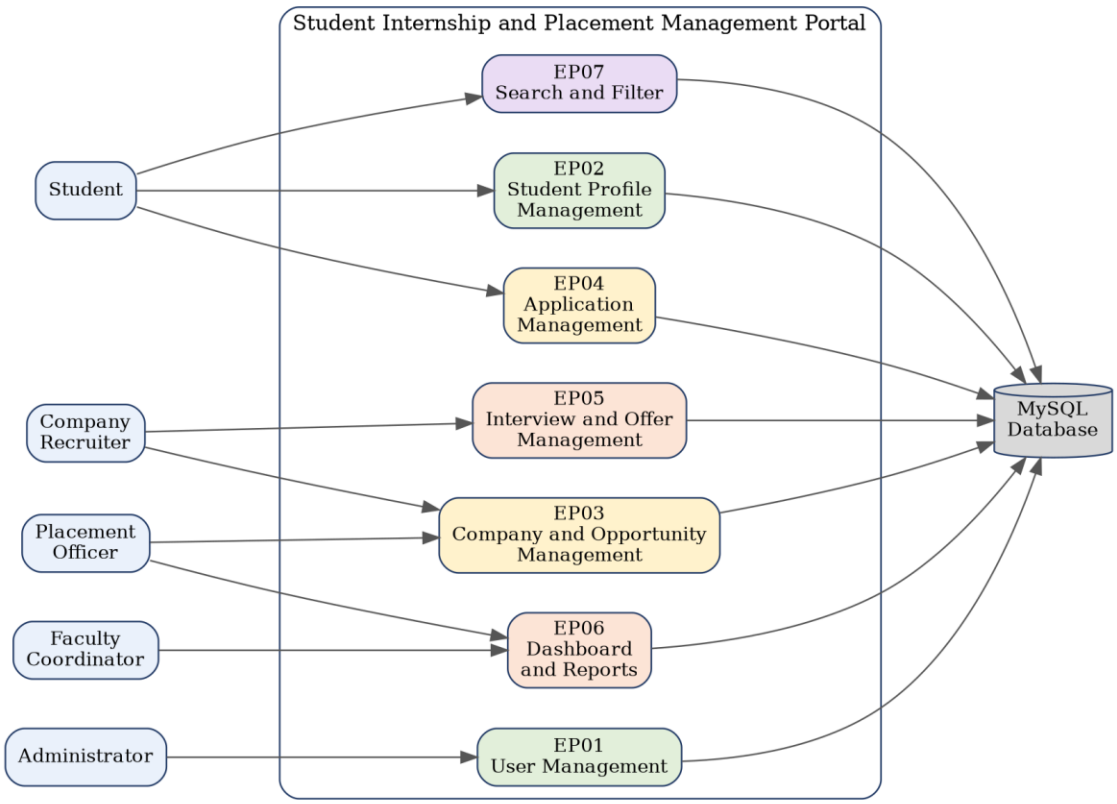


Figure 1: System context diagram – user roles, modules (epics) and database

7. Epics and Tasks of the System

The system is organised into seven epics. Each epic contains four tasks, giving 28 tasks (T01–T28) in total. The same task numbers are used in Requirements Gathering, User Stories, Database Queries, the UI design and the backend development.

Epic	Task	Permission	Precondition
EP01 – User Management	T01 – Register User	Public	User is not already registered
EP01 – User Management	T02 – User Login	Registered User	A valid, active account exists
EP01 – User Management	T03 – User Logout	Logged-in User	User is authenticated
EP01 – User Management	T04 – Manage User Access	Administrator / Placement Officer	Target account exists
EP02 – Student Profile Management	T05 – Create Student Profile	Student	Student account is active
EP02 – Student Profile Management	T06 – View Student Profile	Authorized User	Student profile exists

Epic	Task	Permission	Precondition
EP02 – Student Profile Management	T07 – Edit Student Profile	Student / Placement Officer	Student profile exists
EP02 – Student Profile Management	T08 – Upload and Manage Resume	Student	Student profile exists
EP03 – Company and Opportunity Management	T09 – Create Company Profile	Company Recruiter	Recruiter account is approved and active
EP03 – Company and Opportunity Management	T10 – Post Opportunity	Company Recruiter / Placement Officer	Company profile exists
EP03 – Company and Opportunity Management	T11 – Edit Opportunity	Company Recruiter / Placement Officer	Opportunity exists
EP03 – Company and Opportunity Management	T12 – Close or Delete Opportunity	Company Recruiter / Placement Officer / Administrator	Opportunity exists
EP04 – Application Management	T13 – Apply for Opportunity	Student	Opportunity is open and the student is eligible
EP04 – Application Management	T14 – View Application Details	Authorized User	Application exists
EP04 – Application Management	T15 – Update Application Status	Company Recruiter / Placement Officer	Application exists
EP04 – Application Management	T16 – Withdraw Application	Student / Placement Officer	Application exists and is not finalized
EP05 – Interview and Offer Management	T17 – Schedule Interview	Company Recruiter / Placement Officer	Application is shortlisted
EP05 – Interview and Offer Management	T18 – View Interview Schedule	Authorized User	Interview is scheduled
EP05 – Interview and Offer Management	T19 – Release Offer	Company Recruiter / Placement Officer	Application is selected
EP05 – Interview and Offer Management	T20 – Respond to Offer	Student	Offer is released and pending
EP06 – Dashboard and Reports	T21 – View Dashboard	Logged-in User	User is authenticated

Epic	Task	Permission	Precondition
EP06 – Dashboard and Reports	T22 – View Opportunity Summary	Authorized User	Opportunity records exist
EP06 – Dashboard and Reports	T23 – View Placement Summary	Authorized User	Application and offer records exist
EP06 – Dashboard and Reports	T24 – Generate Reports	Placement Officer / Faculty Coordinator / Administrator	Records exist for the selected period
EP07 – Search and Filter	T25 – Search Opportunities	Logged-in User	Opportunity records exist
EP07 – Search and Filter	T26 – Filter Opportunities	Logged-in User	Opportunity records exist
EP07 – Search and Filter	T27 – Search Student Records	Company Recruiter / Placement Officer / Faculty Coordinator / Administrator	Student records exist
EP07 – Search and Filter	T28 – Filter Applications	Authorized User	Application records exist

8. Proposed System Overview and Key Features

Feature Area	Description
Secure access	Registration, login, logout, session management, account approval, activation, deactivation and role assignment with complete access history.
Student profile and resume	Academic and personal details (branch, semester, CGPA, skills) with one resume that can be updated and reused for every application.
Opportunity management	Companies post internships and placements with eligibility rules, compensation and deadlines; edits and closures are tracked.
Eligibility-based application	The system checks CGPA, branch, semester, deadline and duplicate applications before accepting an application.
Status tracking	Every change in application status is saved in a history so students, recruiters and the T&P cell see the same information.
Interviews and offers	Interview date, time, mode and venue/link are published to the student; offers are released and answered online.

Feature Area	Description
Dashboards and reports	Role-specific dashboards and branch-wise, company-wise and period-wise reports in exportable form.
Search and filter	Keyword search and combined filters for opportunities, students and applications.

9. System Architecture and Technology Stack

The portal follows a three-tier architecture. The presentation layer is a responsive web application whose screens are first designed in Figma. The business layer is a Spring Boot application that exposes REST APIs and applies the business rules. The data layer is a MySQL database accessed through Spring Data JPA. All APIs are tested using Postman.

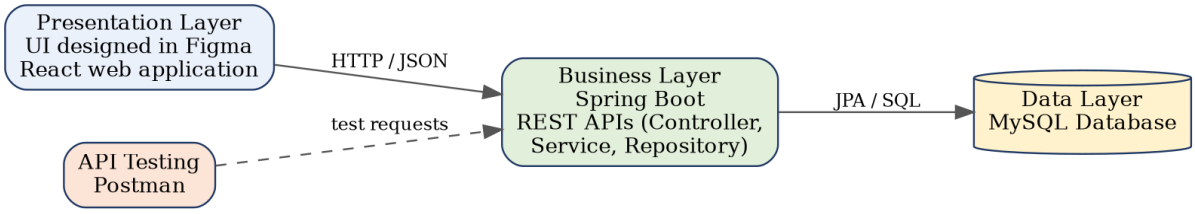


Figure 2: High-level architecture of the portal

Layer	Technology	Purpose
UI Design	Figma	Wireframes, high-fidelity screens, components and prototype (IA 7–8).
Frontend	React, HTML, CSS, JavaScript	Responsive user interface for all roles.
Backend	Java 17, Spring Boot, Spring Data JPA	Business logic, validation and REST APIs (IA 10–11).
Database	MySQL 8.0	Persistent storage of users, profiles, opportunities, applications, interviews, offers and history (IA 5–6).
Testing	Postman	API testing for positive, negative and validation cases (IA 12).
Tools	Git and GitHub, VS Code / IntelliJ IDEA, Maven	Version control, development and build.

10. Assumptions, Constraints and Dependencies

Assumptions

- Students, recruiters and staff have a valid e-mail address, a phone number and internet access.
- Users have basic knowledge of using web applications.
- The Placement Officer verifies the academic details (such as CGPA) entered by students.
- Companies visiting the college will use the portal for their recruitment activities.

Constraints

- The project must be completed within the semester using open-source tools.
- Internet connectivity is required for all real-time operations.
- The first version supports a single college and web browsers only.

Dependencies

- Each assignment depends on the previous one: requirements depend on this initiation, user stories on the requirements, the database on the user stories, the UI and backend on the database design, and testing on the backend.
- Availability of MySQL, JDK, Node.js, Figma and Postman on the development machines.

11. Risk Analysis and Mitigation

Risk	Likelihood	Impact	Mitigation Plan
Requirements change during development	Medium	Medium	Maintain a prioritised backlog and trace every change to the affected assignments.
Unauthorised access to student or company data	Low	High	Role-based access, password hashing, input validation and access history.
Entry of wrong academic data by students	Medium	Medium	Verification of CGPA and eligibility details by the Placement Officer.
Low adoption by students and companies	Medium	High	Simple, user-friendly design; orientation session conducted by the T&P cell.
Team lacks skills in a technology	Medium	Medium	Practice sessions, tutorials and guidance from the faculty mentor.
Time shortage before submission	High	High	Follow the project plan and complete high-priority (Must-have) stories first.
Data loss or corruption	Low	High	Regular database backups and version control of all code and documents.

12. Project Plan and Deliverables

No.	Industrial Assignment	Deliverable
1	Project Initiation	Project definition, scope, stakeholders, epics and plan (this document).
2	Requirements Gathering	Gathering techniques, functional and non-functional requirements, business rules and traceability.
3–4	User Stories	28 user stories (EP01–EP07) with permission matrices, acceptance criteria and definitions of done.
5	Creating Database Queries	CREATE TABLE statements and task-wise SQL queries for T01–T28.
6	ER Diagram for the DB Design	Entity inventory, ER diagrams and cardinality table.
7–8	UI Design using Figma	Design system, wireframes, high-fidelity screens and prototype.
9	Running a full stack project	Setup and execution of the database, backend and frontend together.
10–11	Developing Backend code using Spring Boot	Entities, repositories, services, controllers and exception handling.
12	Testing Backend code using Postman	Postman collection, test scripts and test report.

13. Success Criteria and Key Performance Indicators

- A student can register, complete a profile, apply to an eligible opportunity and track the status from start to finish without help from the T&P cell.
- A company can post an opportunity and take applicants through shortlist, interview, selection and offer stages using only the portal.
- The system rejects ineligible, late and duplicate applications every time.
- The Placement Officer can produce branch-wise and company-wise placement reports without manual calculation.
- Every role sees only the information and actions permitted for that role.
- All 28 tasks are covered by user stories, database queries, UI screens, backend APIs and Postman test cases.

14. Expected Benefits

Beneficiary	Benefit
Students	Equal and timely access to opportunities, one profile and resume for all applications, transparent status tracking.
Companies	Quick access to eligible candidates, less paperwork and a clear selection workflow.
Placement Officer	Less manual work, accurate records and instant reports.
Faculty and Management	Reliable placement statistics for planning, accreditation and decision-making.

15. Conclusion

The project initiation has defined the purpose, scope, user roles, seven epics with 28 tasks, technology stack, risks and plan for the Student Internship and Placement Management Portal. This document will be used as the base for Requirements Gathering in Industrial Assignment 2 and for every later assignment.